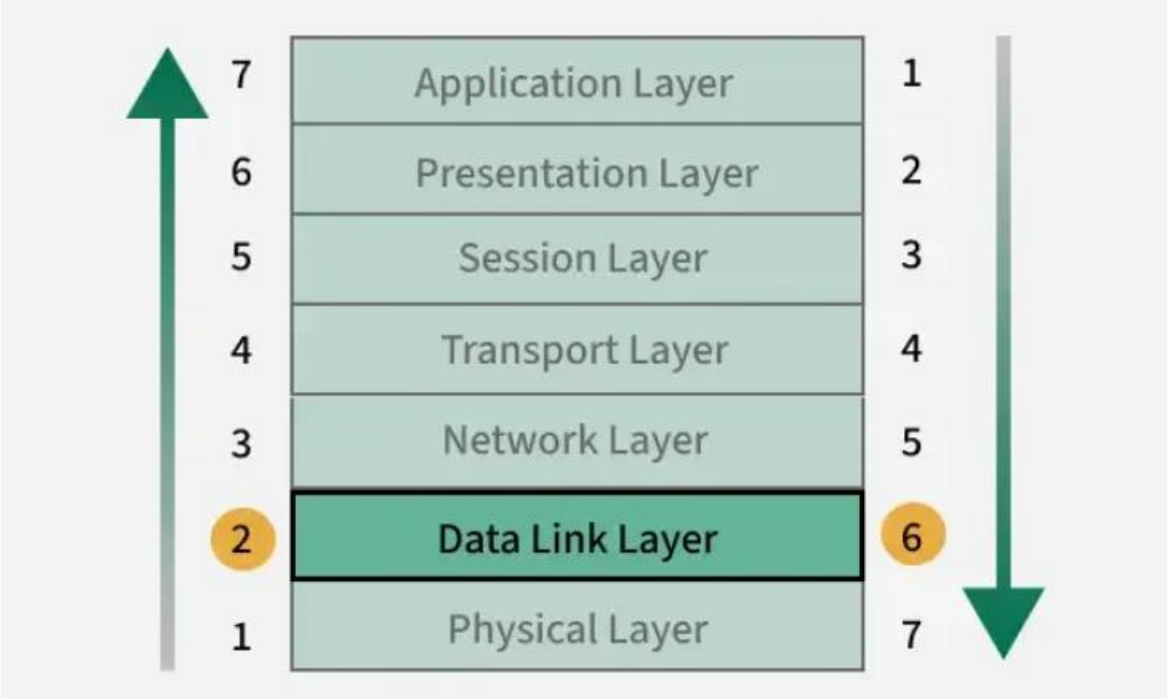
UNIT-II: Data link layer:

- Data link layer: Design issues, Framing: fixed-size framing, variable-size framing, flow control, error control, error detection and correction codes, CRC, Checksum: idea, one's complement internet checksum, services provided to Network Layer, Elementary Data Link Layer protocols: simplex protocol, Simplex stop and wait, Simplex protocol for a noisy channel.
- Sliding window protocol: One bit, go-back-N, selective repeat; stop-and-wait protocol; data link layer in HDLC; point-to-point protocol (PPP).

Data Link Layer in OSI Model

The data link layer is the second layer from the bottom in the OSI (Open System Interconnection) network architecture model.

- ✓ Responsible for the node-to-node delivery of data within the same local network.
- ✓ Major role is to ensure error-free transmission of information.
- ✓ Also responsible for encoding, decoding, and organizing the outgoing and incoming data.
- ✓ Considered as the most complex layer of the OSI model as it hides all the underlying complexities of the hardware from the other above layers.



Sub-Layers of The Data Link Layer

The data link layer is further divided into two sub-layers, which are as follows:

1. **Logical Link Control (LLC):** This sublayer of the data link layer deals with multiplexing, the flow of data among applications and other services, and LLC is responsible for providing error messages and acknowledgments as well.
2. **Media Access Control (MAC):** MAC sublayer manages the device's interaction, responsible for addressing frames, and also controls physical media access. The data link layer receives the information in the form of packets from the Network layer, it divides packets into frames and sends those frames bit-by-bit to the underlying physical layer.

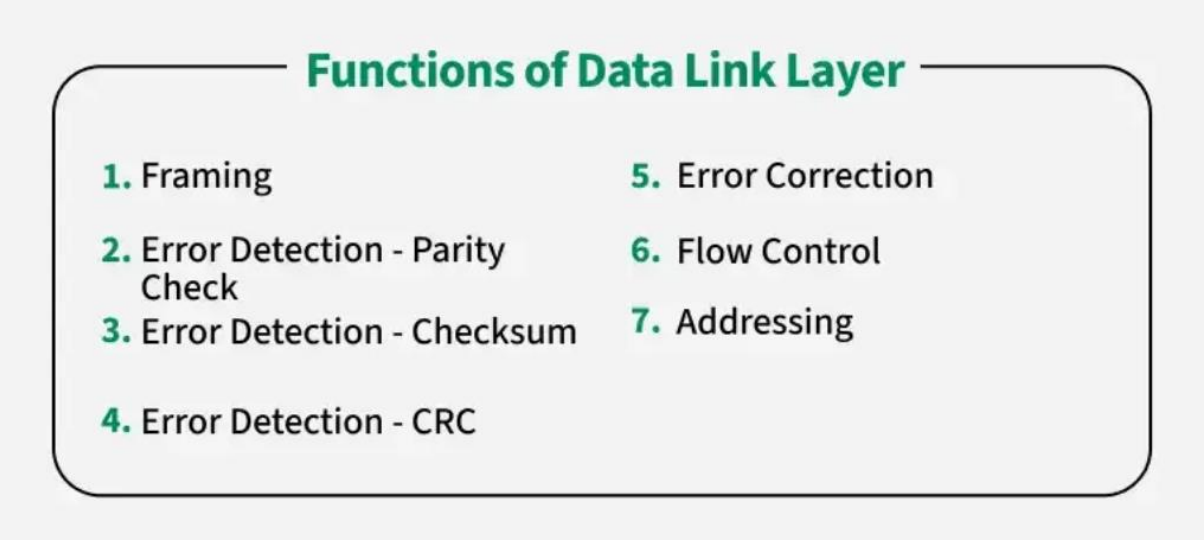


Fig 1

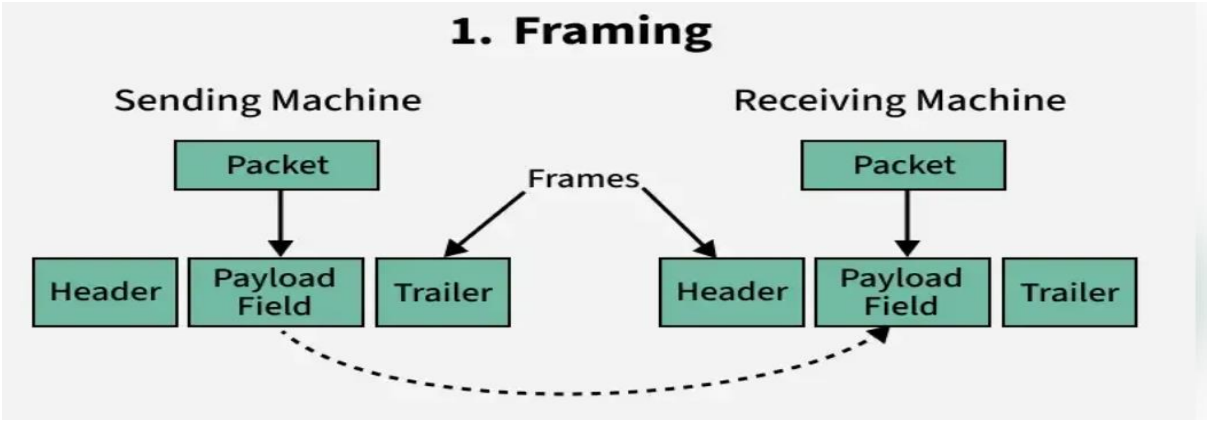


Fig: 2

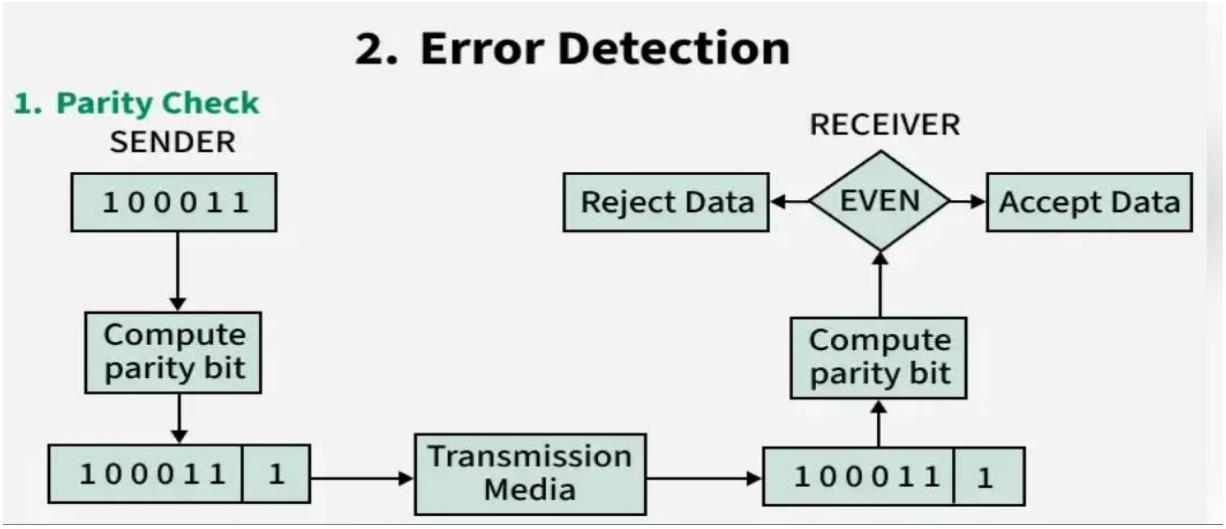


Fig 3

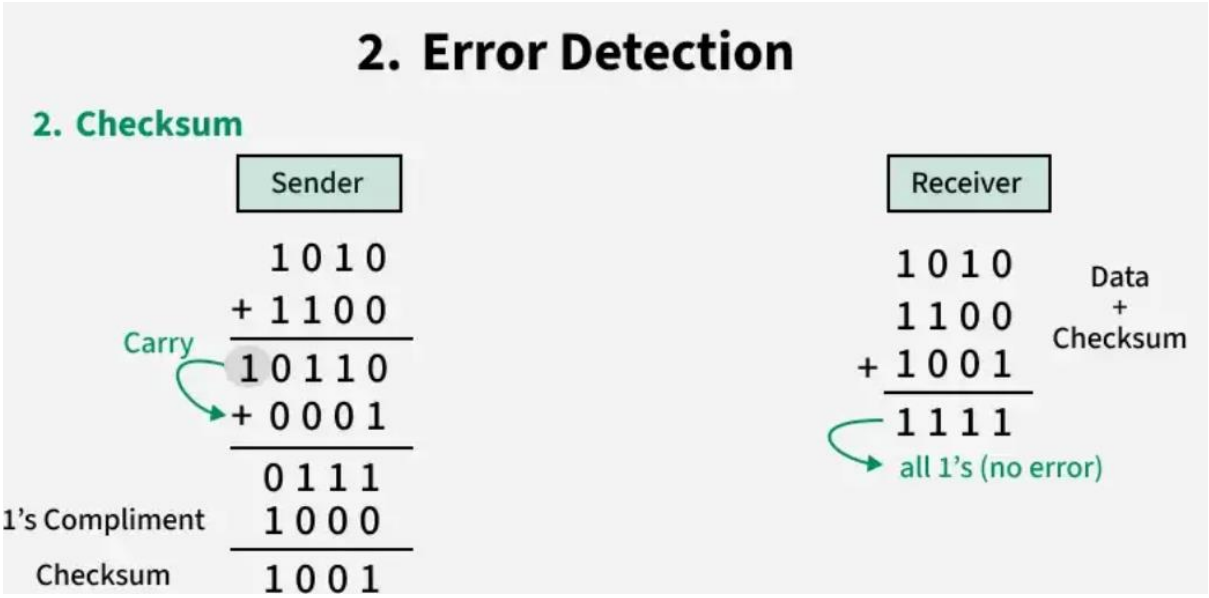


Fig 4

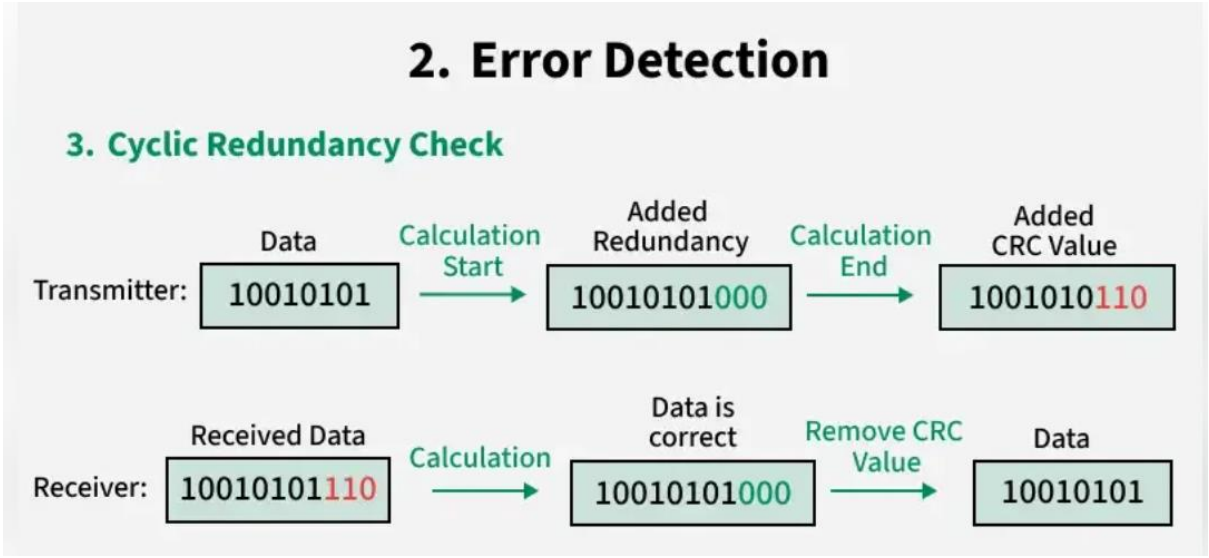


Fig 5

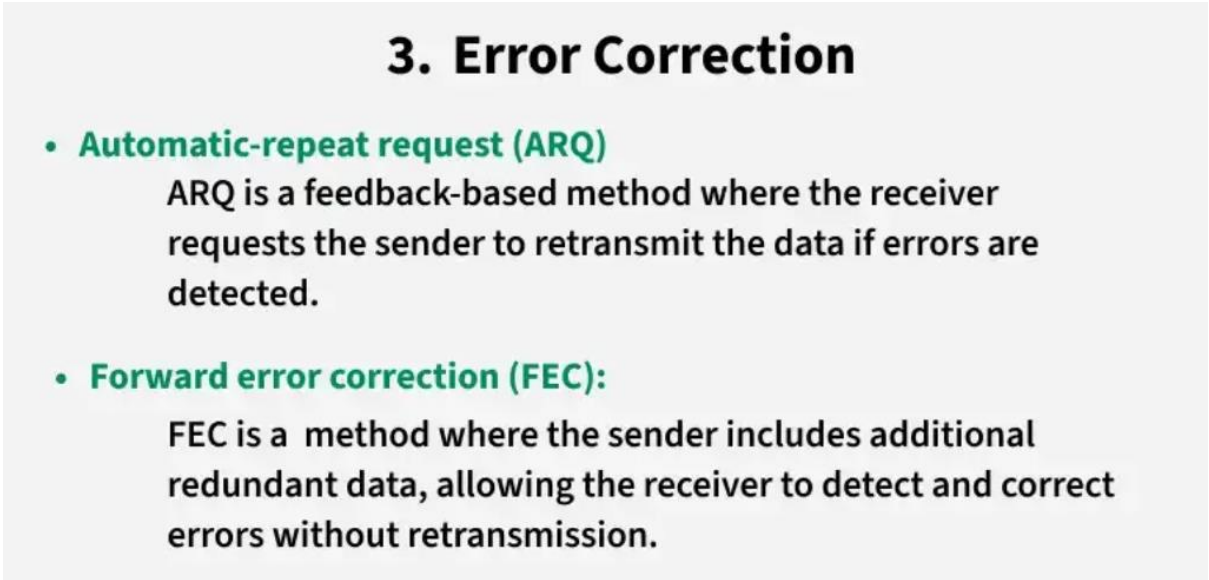


Fig 6

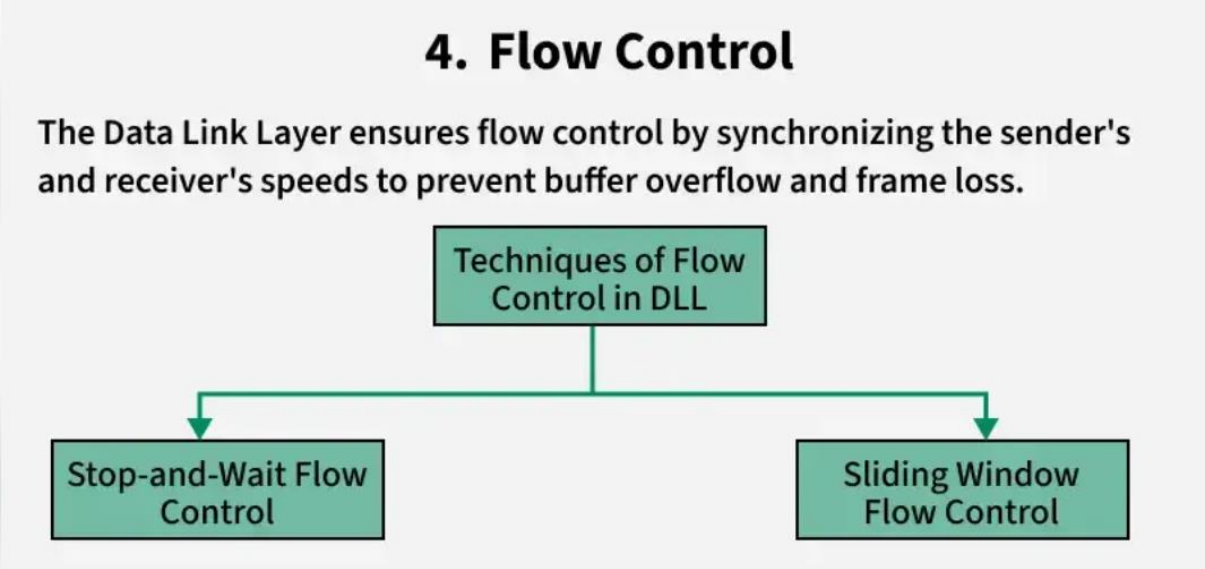


Fig 7

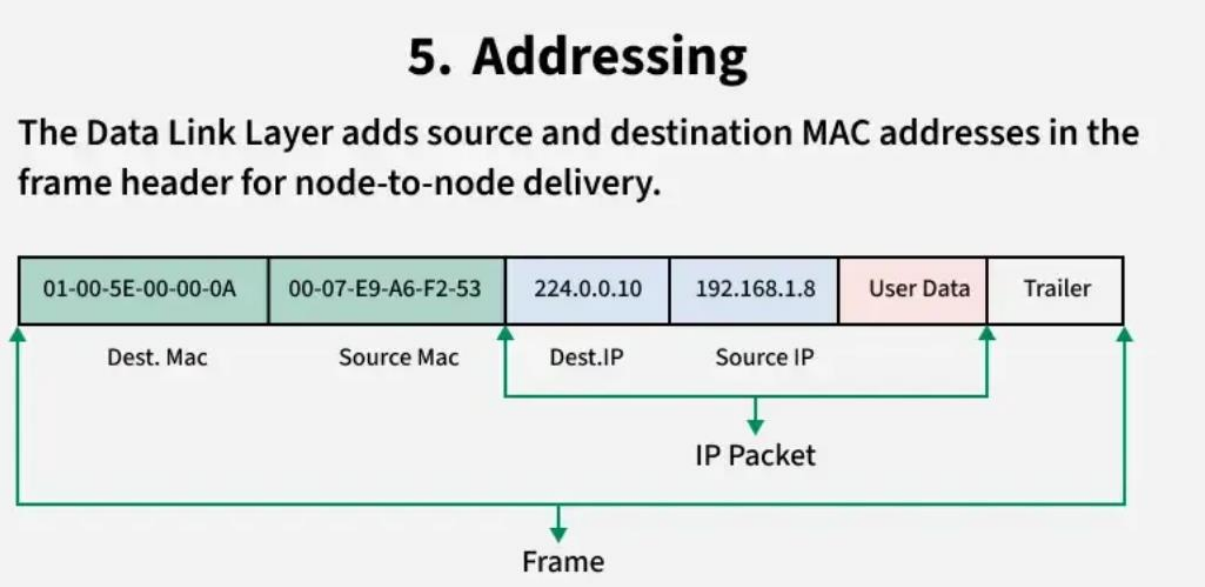
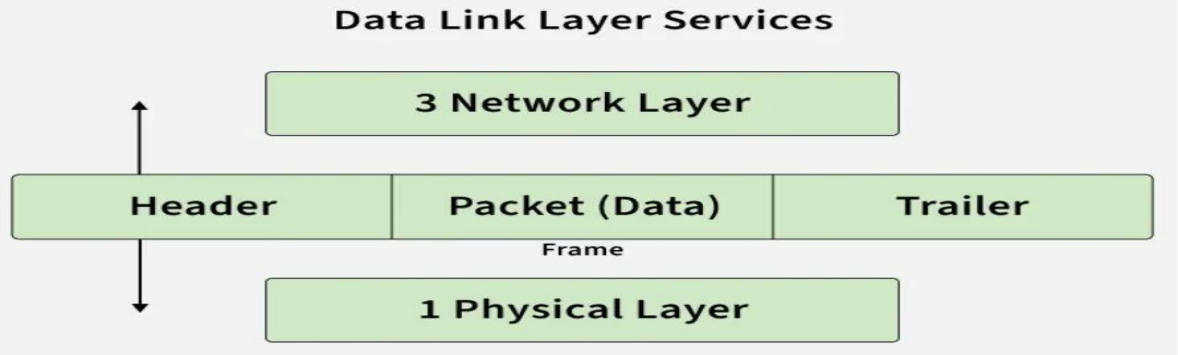


Fig: 8

- Fixed-size framing, variable-size framing

  Framing in Data Link Layer :A frame is the basic unit of data transmission at the Data Link Layer, consisting of a structured group of bits that carries data along with control information.
  - Frames include control information such as addressing and error detection.
  - Framing is performed at the Data Link Layer (DLL) of the OSI model.
  - It divides a continuous stream of bits into manageable and meaningful units.
  - Framing enables reliable transmission and proper processing at the receiver.
  - It helps in synchronization between the sender and receiver.



Purpose of Framing

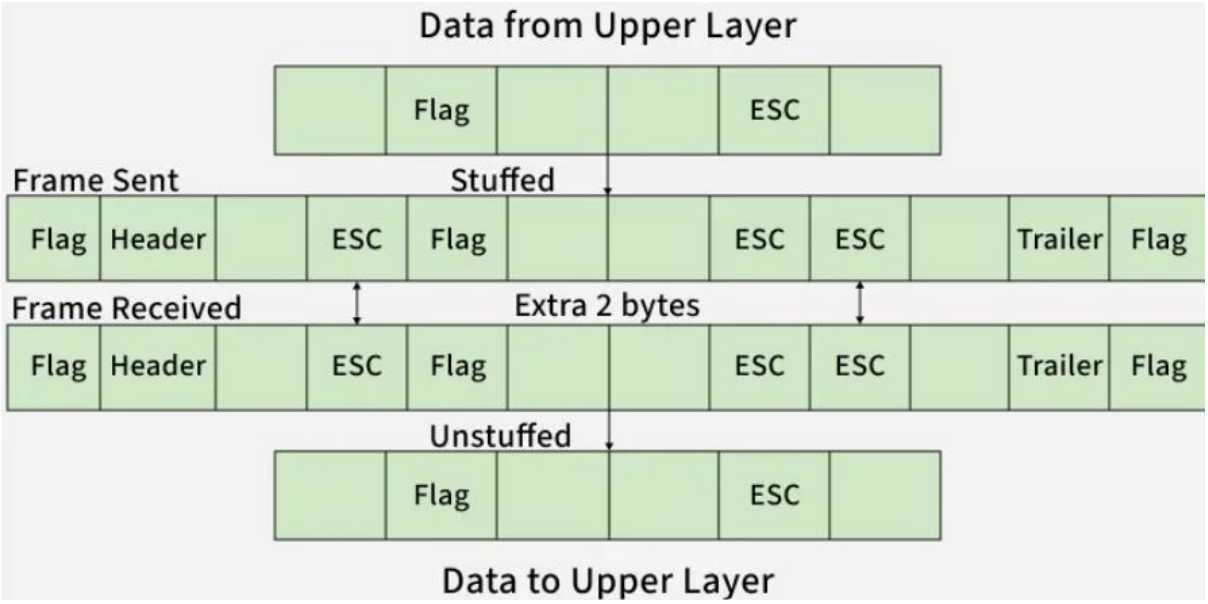
Framing serves several purposes in the Data Link Layer:

- It defines clear frame boundaries so the receiver can correctly identify each frame.
- It carries source and destination addresses to support accurate point-to-point communication.
- It supports error detection and correction to maintain data integrity.
- It ensures reliable and well-organized data transmission.

Types of Framing

1. Fixed-Size Framing

- Frames have a predefined fixed size
- No start or end delimiters are required, as the frame length defines the boundary.
- **Drawback:** Causes internal fragmentation when data is smaller than the frame size, leading to bandwidth wastage.
- **Solution: Padding** is added to fill unused space and complete the frame.



2. Variable-Size Framing

Variable-size framing uses explicit frame boundary indicators to identify the start and end of a frame.

- **Length Field:**

  The frame contains a length field that specifies its size (used in IEEE 802.3 Ethernet). If this field is corrupted, the receiver cannot correctly interpret the frame.
- **End Delimiter (ED):**

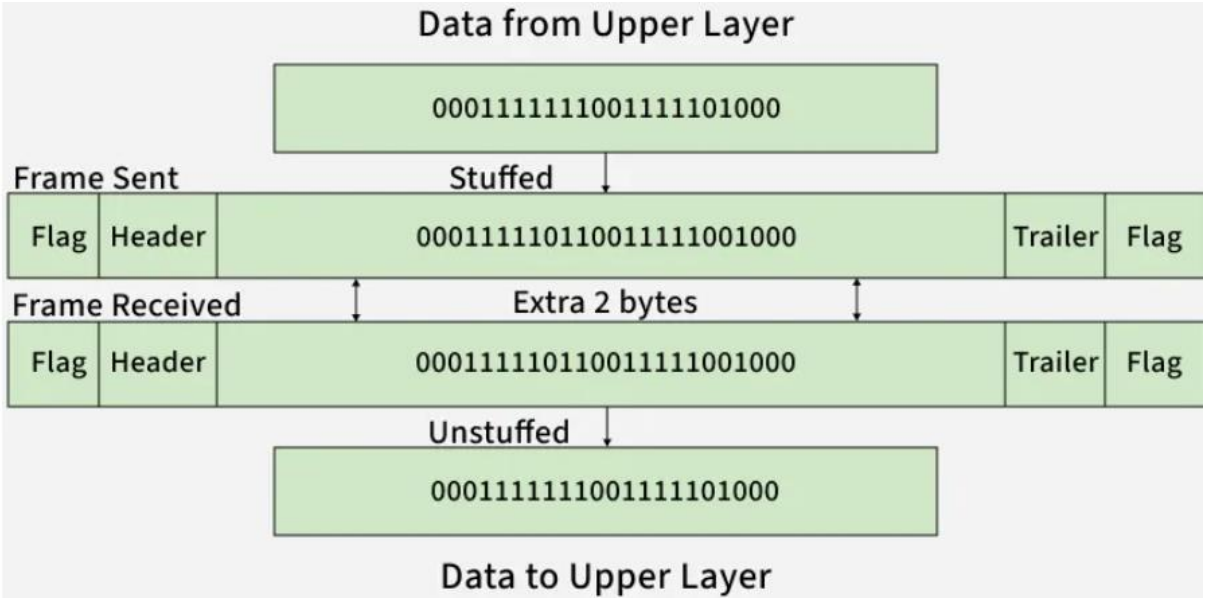
  A unique pattern marks the end of the frame (used in Token Ring). The delimiter pattern may also appear within the data.

To resolve delimiter ambiguity, **stuffing techniques** are applied:

(a) Character/Byte Stuffing

- Used when data is character-based.
- If the data contains the special delimiter character (e.g., $), an escape character (e.g., \O) is inserted to distinguish it.

(b) Bit Stuffing



- ED pattern is a sequence of bits (e.g., 01111).
- Whenever this pattern appears in data, a 0 is inserted to prevent confusion.
  Example Problem: Data = 011100011110, ED = 0111, After bit stuffing -> 011010001101100
  Another Example: Data = 110001001, ED = 1000, After bit stuffing -> 11001010011

Problems in Framing

Framing in computer networks faces several challenges:

- **Start and End Detection:** The receiver must correctly identify frame boundaries using delimiters or flags.
- **Synchronization:** Sender and receiver must stay aligned on frame timing, especially in high-speed networks.
- **Error Handling:** Noise can corrupt data or delimiters; error detection methods like CRC or checksums are used.
- **Overhead:** Headers and trailers add control information but reduce usable bandwidth.
- **Efficiency:** The goal is to maximize payload while minimizing overhead and processing delays.
- **Compatibility:** Different devices may use different framing methods, causing misinterpretation.

➢ Flow control & Error control:

In data transmission, the general aspect of maintaining effective and accurate transfer of data between gadgets is very important. This is where the flow control and the error control mechanisms will be of help. Flow control regulates the amount of data being transmitted from the sender to the receiver so that it does not overpower the receiver, while error control concerns itself with errors

that may occur in the transmission process. Protocols are two categories that run again in the Data Link Layer to ensure that the transmission of data is accurate and efficient.

What is Flow Control?

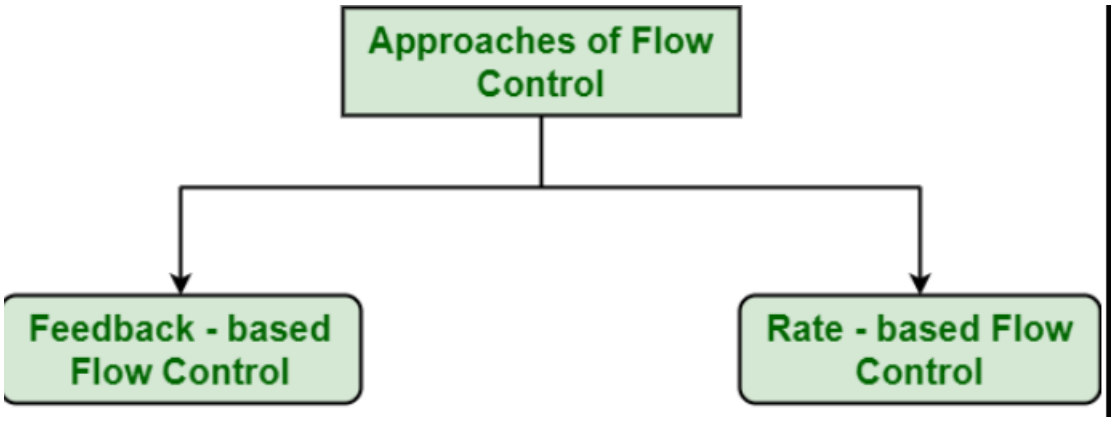
It is an important function of the Data Link Layer. It refers to a set of procedures that tells the sender how much data it can transmit before waiting for acknowledgment from the receiver.

Purpose of Flow Control

Any receiving device has a limited speed at which it can process incoming data and also a limited amount of memory to store incoming data. If the source is sending the data at a faster rate than the capacity of the receiver, there is a possibility of the receiver being swamped. The receiver will keep losing some of the frames simply because they are arriving too quickly and the buffer is also getting filled up.

This will generate waste frames on the network. Therefore, the receiving device must have some mechanism to inform the sender to send fewer frames or stop transmission temporarily. In this way, flow control will control the rate of frame transmission to a value that can be handled by the receiver.

Approaches to Flow Control



Example -Stop & Wait Protocol

1. Feedback-Based Flow Control

In this approach, the receiver sends explicit feedback to the sender about its ability to receive data. Based on this feedback, the sender adjusts its transmission rate to match the receiver's current capacity.

- The sender transmits data only after receiving acknowledgments or window updates from the receiver
- Transmission rate is dynamically adjusted according to receiver or network conditions
- Commonly used in protocols like **TCP** for efficient and reliable communication

Techniques in Computer Architecture

Ready-Signal Handshaking

- This technique is used between components operating at different speeds or clock domains
- The receiver sends a ready signal indicating it can accept data
- Data transfer occurs only when the receiver is ready, preventing data loss

Credit-Based flow control

- This technique is used in high-speed protocols such as **Fibre Channel**
- The receiver provides credits to the sender, each representing available buffer space
- The sender can transmit one frame per credit
- Credits are restored as the receiver processes data
- More efficient than simple stop-and-wait flow control

2. Rate-Based Flow Control

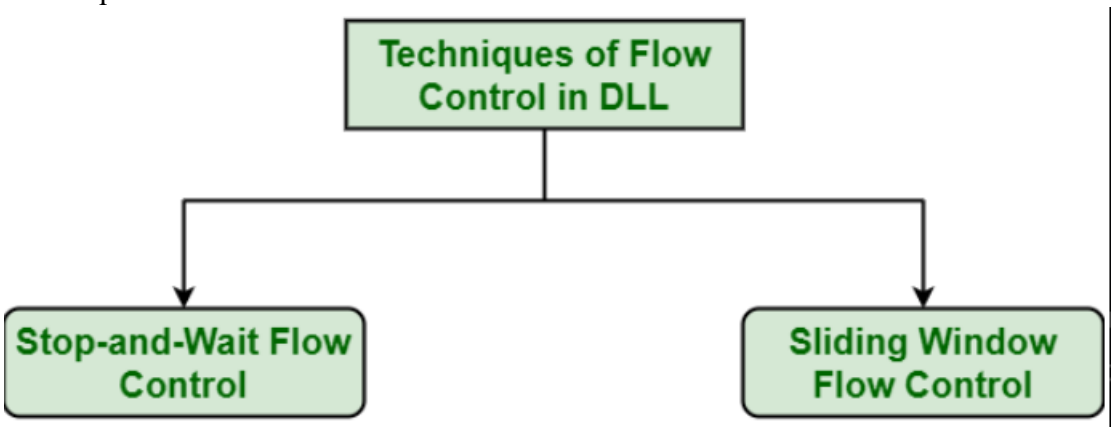
In rate-based flow control, the sender transmits data at a fixed or pre-negotiated rate without receiving continuous feedback from the receiver. This approach assumes that the receiver can process data at the agreed transmission rate.

- Sender controls the transmission rate independently
- Does not rely on acknowledgments or window updates
- Suitable for predictable and time-sensitive data transmission
- Commonly used in streaming, multimedia, and hardware systems

Techniques in computer architecture

- **Leaky Bucket Algorithm:** Maintains a constant data output rate by smoothing bursty input traffic, ensuring steady and controlled transmission.
- **Token Bucket Algorithm:** Generates tokens at a fixed rate, allowing data transmission only when tokens are available. A larger bucket size supports short bursts while preserving the average transmission rate.
- **Traffic Shaping and Policing:** Regulate network traffic based on a defined rate. Traffic shaping delays excess packets to smooth traffic flow, while policing drops packets that exceed the allowed rate.

➢ Techniques of Flow Control



1. Stop-and-Wait Flow Control

Stop-and-Wait is a simple feedback-based flow control technique in which the sender transmits one frame at a time and waits for an acknowledgment from the receiver before sending the next frame. If the acknowledgment is not received within a specified timeout period, the sender retransmits the frame to ensure reliable delivery.

- Simple and easy to implement
- Ensures reliable data transmission
- Poor link utilization in high-latency or high-bandwidth networks due to idle waiting

2. Sliding Window Flow Control

Sliding Window is a feedback-based flow control technique that allows the sender to transmit multiple frames before receiving acknowledgments. The sender maintains a *window* of unacknowledged frames and moves (slides) the window forward as acknowledgments are received, enabling continuous and pipelined data transmission.

- Improves bandwidth utilization by enabling pipelined transmission.
- Allows multiple frames to be in transit simultaneously
- Requires additional memory and complexity for window and buffer management

The significance of flow control is multifaceted:

- **Prevents Data Loss:** Regulates data transmission to avoid buffer overflow at the receiver
- **Improves Throughput:** Optimizes the use of network resources without overloading any component

- **Maintains System Stability:** Ensures smooth operation of hardware and software components by preventing excessive data input
- **Ensures Fairness:** Supports balanced data transmission across multiple connections in shared environments such as switches and routers

Importance of Flow Control

Flow control plays a crucial role in efficient and reliable data communication:

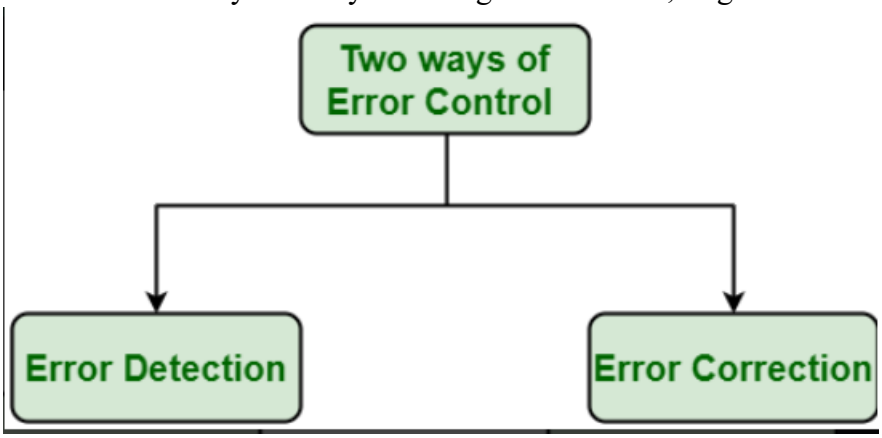
- **Prevents Data Loss:** Controls the transmission rate to avoid buffer overflow at the receiver
- **Improves Throughput:** Ensures efficient utilization of network resources without overloading any device
- **Maintains System Stability:** Prevents hardware and software components from stalling or failing due to excessive data input
- **Ensures Fairness:** Enables balanced data transmission across multiple connections in shared environments such as switches and routers
- **What is Error Control?**
- The error control function of the data link layer detects the errors in transmitted frames and re-transmits all the erroneous frames.
- **Purpose of Error Control**
- The function of error control function of the data link layer helps in dealing with data frames that are damaged in transit, data frames lost in transit and acknowledged frames that are lost in transmission. The method used for error control is called Automatic Repeat Request (ARQ) which is used for the noisy channel.
- **Example -**Stop & Wait ARQ and Sliding Window ARQ

Error control in the Data Link Layer ensures that frames transmitted by the sender are received correctly and reliably by the receiver. It focuses on:

- Detecting errors that occur during data transmission
- Correcting errors through retransmission when required
- Ensuring reliable frame delivery over a noisy communication channel

The primary mechanism used for error control is Automatic Repeat Request (ARQ). In ARQ, if an error is detected or an ACK is not received, the sender retransmits the affected frame(s) until they are successfully received.

There are basically two ways of doing Error control, as given below :



- **Error Detection:** Error detection refers to the identification of errors in transmitted data. These errors may occur due to noise or other transmission impairments while data is being sent from the transmitter to the receiver in a communication system. It involves techniques used to detect corrupted, distorted, or unclear data before it is processed further.
- **Error Correction:** Error correction refers to the process of fixing detected errors to recover the original, error-free data. It involves reconstructing the correct data either by using additional information or through retransmission. However, error correction techniques are

generally complex, costly, and resource-intensive, which is why they are used selectively.

1. Stop-and-Wait ARQ: In Stop-and-Wait ARQ, the sender transmits one data frame at a time and waits for an ACK (Acknowledgment) from the receiver. If the ACK is not received within a time limit, the sender retransmits the frame. Once the ACK is received, the sender sends the next frame. This process continues until all frames are successfully delivered.

2. Sliding Window ARQ: Sliding Window ARQ is used for continuous data transmission, allowing the sender to transmit multiple frames without waiting for an acknowledgment for each one. This improves channel utilization and overall efficiency. Sliding Window ARQ is mainly divided into two types:

Techniques of error control

Stop-and-Wait ARQ:

In Stop-and-Wait ARQ, the sender transmits one data frame at a time and waits for an ACK (Acknowledgment) from the receiver. If the ACK is not received within a time limit, the sender retransmits the frame. Once the ACK is received, the sender sends the next frame. This process continues until all frames are successfully delivered.

Sliding Window ARQ:

Sliding Window ARQ is used for continuous data transmission, allowing the sender to transmit multiple frames without waiting for an acknowledgment for each one. This improves channel utilization and overall efficiency. Sliding Window ARQ is mainly divided into two types:

Go-Back-N ARQ:

It is an error control protocol in which the sender is allowed to send multiple frames (up to a fixed window size) without waiting for ACKs.

- The receiver sends ACKs for correctly received frames.
- If a frame is lost or arrives with an error, the receiver discards that frame and all subsequent frames.
- The sender then retransmits the erroneous frame along with all the frames sent after it.
- When no errors occur, the protocol behaves like a normal sliding window mechanism.

This method is simple to implement but may cause unnecessary retransmissions.

Selective Repeat ARQ:

It is an error control protocol in which only the frames that are lost or corrupted are retransmitted, unlike Go-Back-N where multiple frames are resent.

- It improves efficiency by avoiding unnecessary retransmissions.
- The sender retransmits only those frames for which a NAK is received or a timeout occurs.
- Each frame is acknowledged individually, allowing correctly received frames to be accepted and stored.
- Due to the need for additional buffering and tracking, the protocol is more complex, which limits its use compared to simpler ARQ schemes.

The main difference between Go-Back-N ARQ and Selective Repeat ARQ is in the retransmission strategy. In Go-Back-N ARQ, if a frame is lost or corrupted, the sender retransmits that frame along with all subsequent frames in the window. In contrast, Selective Repeat ARQ retransmits only the specific frame that is lost or damaged, making it more efficient.

- **Error Detection in Computer Networks**

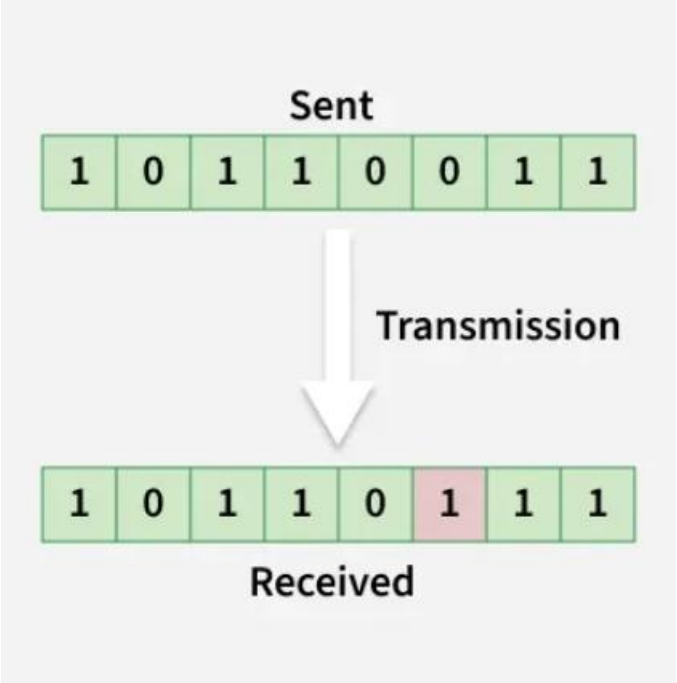
Error detection is a technique used in computer networks to identify whether transmitted data has been corrupted during communication. The sender adds redundant bits to the data before transmission, and the receiver verifies the integrity of the received frame by checking these bits. If an error is detected, the corrupted frame is discarded, and retransmission may be requested through appropriate error control mechanisms.

- Detects errors caused by noise, interference, or signal distortion
- Relies on redundant bits for consistency checking
- Common techniques include Parity Check, Checksum, and Cyclic Redundancy Check (CRC)
- Helps maintain data integrity and reliable communication

Types of Errors

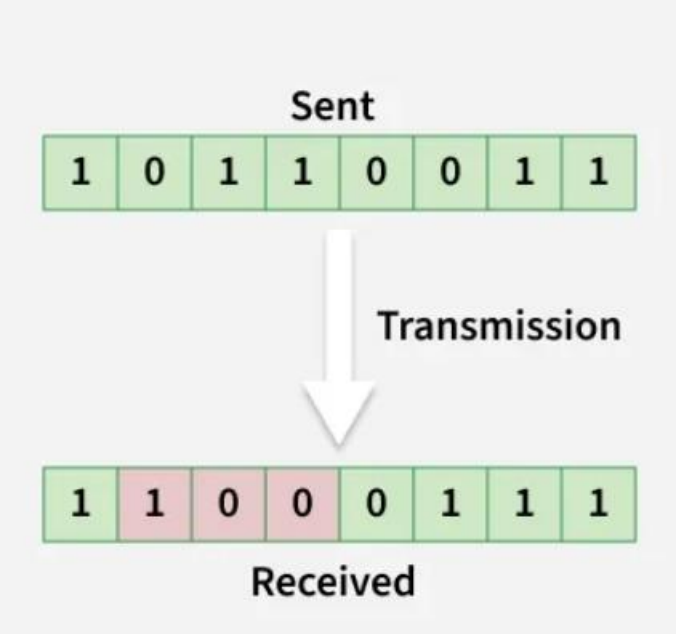
1. Single-Bit Error

A single-bit error occurs when only one bit of the transmitted data unit is altered during transmission, resulting in corrupted data.



2. Burst Error

A burst error occurs when two or more consecutive bits in a data unit are corrupted during transmission.



Error Detection Methods

To detect errors, a common technique is to introduce redundancy bits that provide additional information. Various techniques for error detection include:

Simple Parity Check
Two-Dimensional Parity Check
Checksum
Cyclic Redundancy Check (CRC)

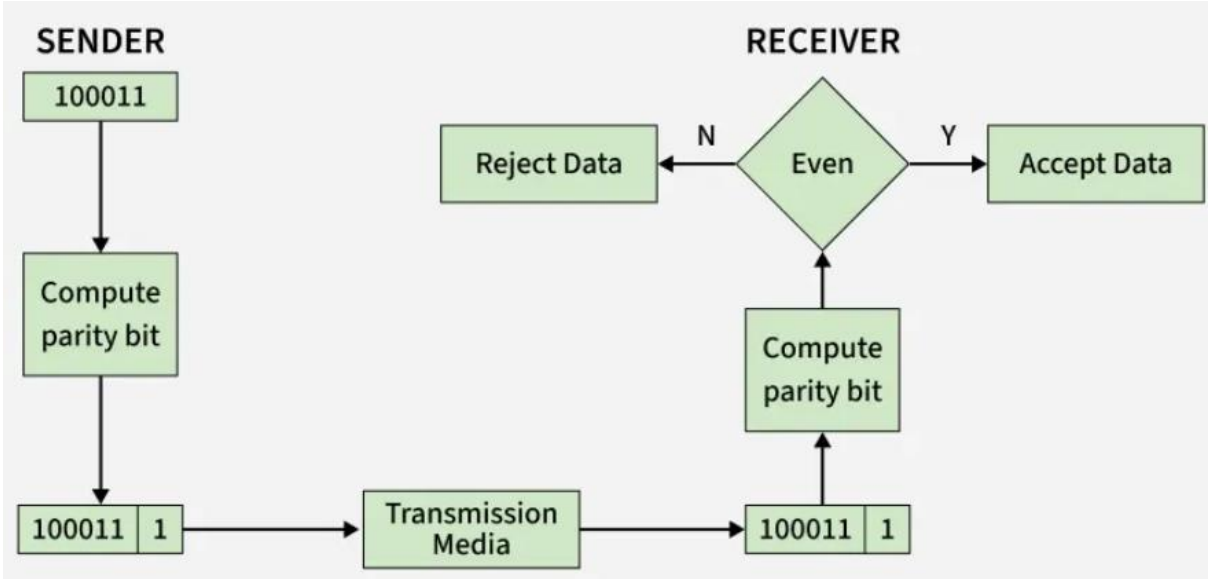
1. Simple Parity Check

Simple bit parity is a basic error detection technique in which an extra bit, called a parity bit, is added to a data unit before transmission. This parity bit helps the receiver determine whether the transmitted data has been corrupted.

- **Detects all single-bit errors in transmitted data.**
- **Detects any odd number of bit errors, since odd bit changes alter the parity.**
- **Easy to implement in both hardware and software.**
- **Fails to detect errors when an even number of bits are corrupted.**
- **Not suitable for noisy communication channels with frequent errors.**

Parity bits are used to maintain a specific parity condition in the data. There are two types of parity:

- **Even Parity: The parity bit is set so that the total number of 1s in the data (including the parity bit) is even.**
- **Odd Parity: The parity bit is set so that the total number of 1s in the data (including the parity bit) is odd.**

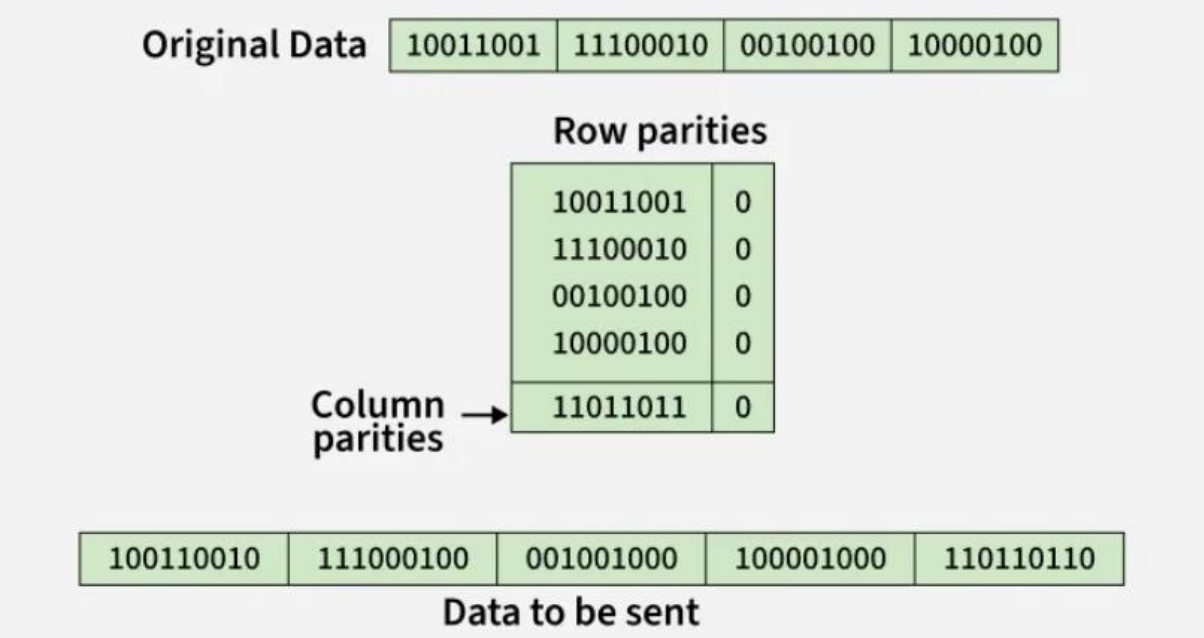


This scheme makes the total number of 1's even, that is why it is called even parity checking.

2. Two-Dimensional Parity Check

In two-dimensional parity check, parity bits are calculated for each row, similar to a simple parity check. In addition, parity bits are also computed for each column. These row and column parity bits are transmitted along with the data. At the receiver, parity bits are recalculated and compared with the received parity bits to detect errors.

- Can detect and correct all single-bit errors by identifying the exact row and column where the error occurred.
- Can detect many multiple-bit errors occurring at different positions in the data matrix.
- Provides better error detection capability than simple parity check.
- Certain patterns of multiple-bit errors may remain undetected.
- If parity bits themselves are corrupted, error detection may fail.



3. Checksum

Checksum is an error detection technique used to detect errors in transmitted data. In this method, the sender divides the data into fixed-size segments and computes a checksum using 1's complement arithmetic. The computed checksum is transmitted along with the data. At the receiver, the same computation is performed to verify data integrity.

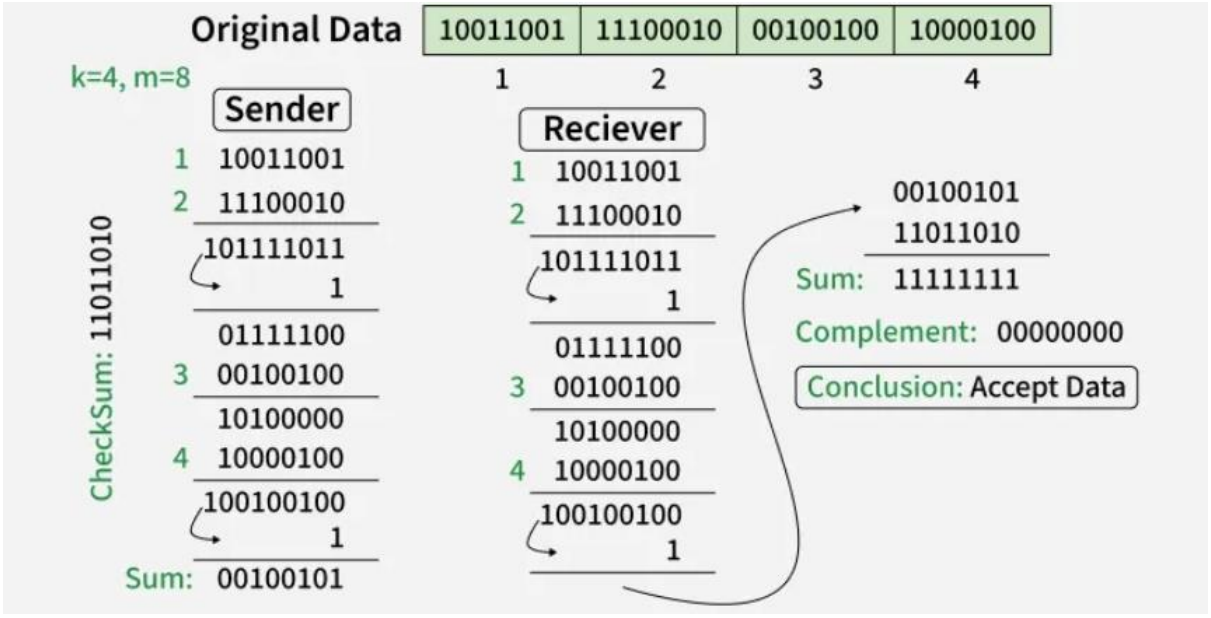
- Can detect many types of errors, including single-bit and multiple-bit errors.
- Provides better error detection than simple parity and two-dimensional parity checks.
- Widely used in networking protocols such as IP, TCP, and UDP.
- Fast to compute and easy to implement in software.
- Supports only error detection and does not provide error correction.
- Some error patterns may go undetected, especially when errors cancel each other during addition.
- Less reliable compared to advanced techniques like Cyclic Redundancy Check (CRC).

Checksum - Operation at Sender's Side

- The data is divided into k segments, each of m bits.
- All segments are added using 1's complement arithmetic.
- The 1's complement of the final sum is calculated to generate the checksum.
- The checksum is appended to the data and transmitted to the receiver.

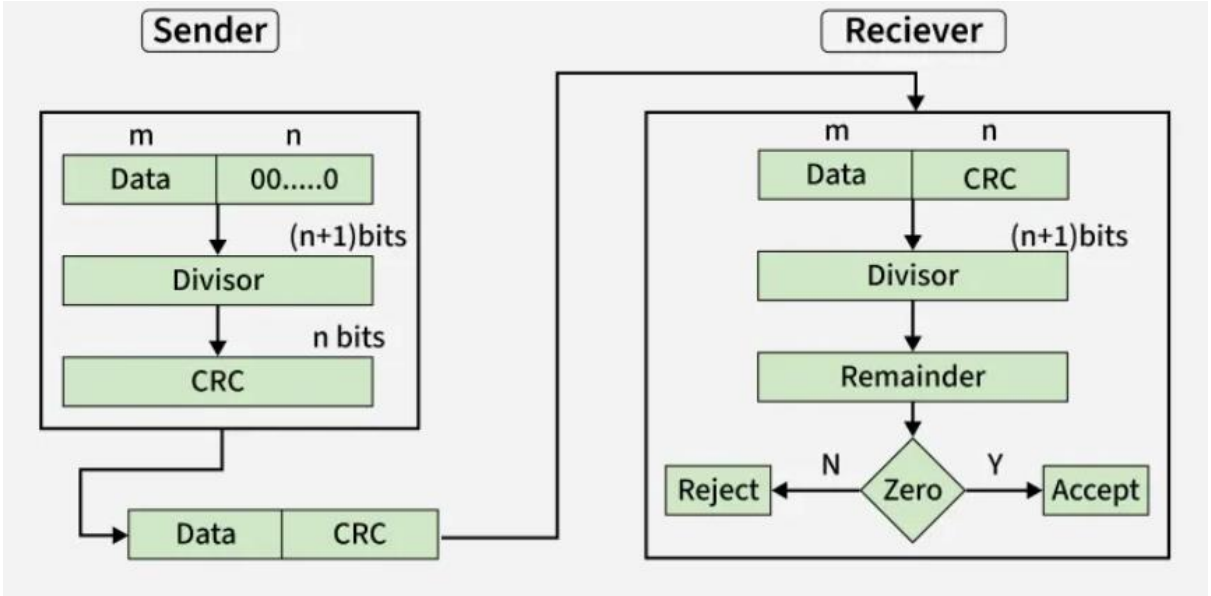
Checksum - Operation at Receiver's Side

- All received segments, including the checksum, are added using 1's complement arithmetic.
- If the final result is all 1s, the data is considered error-free..
- Otherwise, the data is discarded, indicating that an error has occurred.



4. Cyclic Redundancy Check (CRC)

- CRC is based on binary division, unlike checksum which uses addition; redundant CRC bits are appended so the transmitted data becomes exactly divisible by a predefined generator polynomial.
- At the receiver, the data is divided using the same polynomial; a zero remainder means the data is accepted, while a non-zero remainder indicates transmission errors.
- Highly effective in detecting single-bit, multiple-bit, and burst errors, making it more reliable than parity checks and checksum methods.
- Widely used in communication protocols such as Ethernet, HDLC, and USB due to its strong error detection capability.
- Provides only error detection, not error correction; its effectiveness depends on the chosen generator polynomial.
- More computationally complex than simple parity or checksum methods and introduces additional processing overhead.



CRC Working

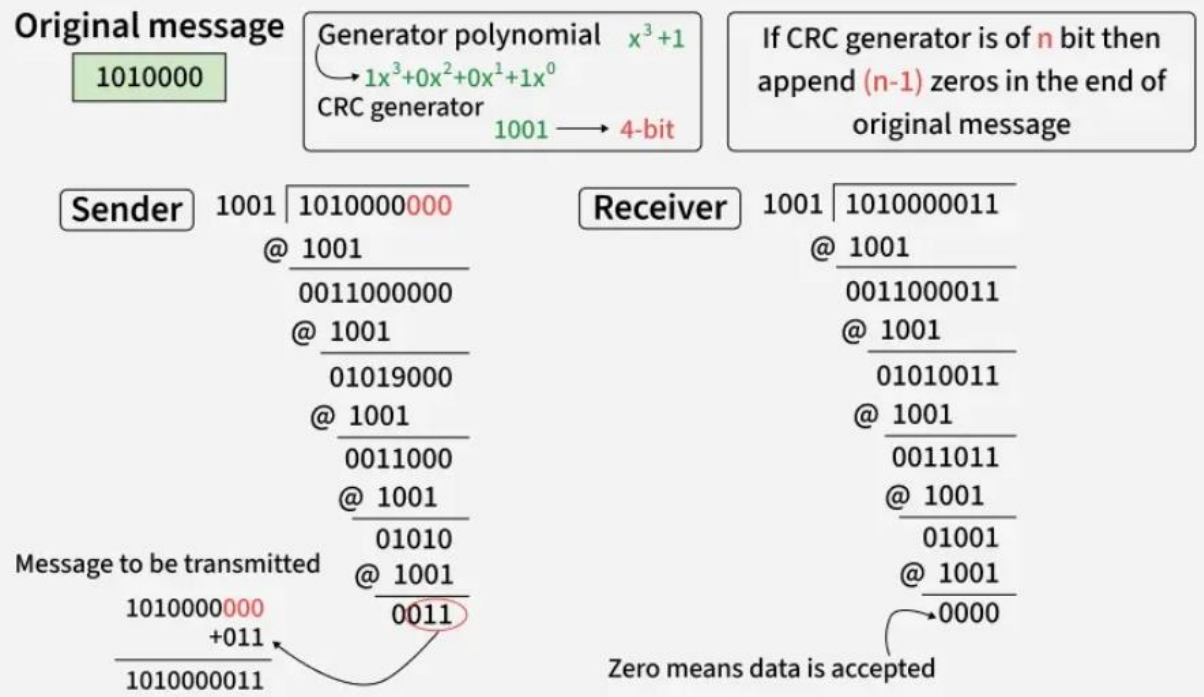
We have given dataword of length n and divisor of length k.

Step 1: Append (k-1) zero's to the original message
Step 2: Perform modulo 2 division
Step 3: Remainder of division = CRC
Step 4: Code word = Data with append k-1 zero's + CRC

Note:

CRC must be k-1 bits
Length of Code word = n+k-1 bits

Example: Let's data to be send is 1010000 and divisor in the form of polynomial is x^3+1. CRC method discussed below.



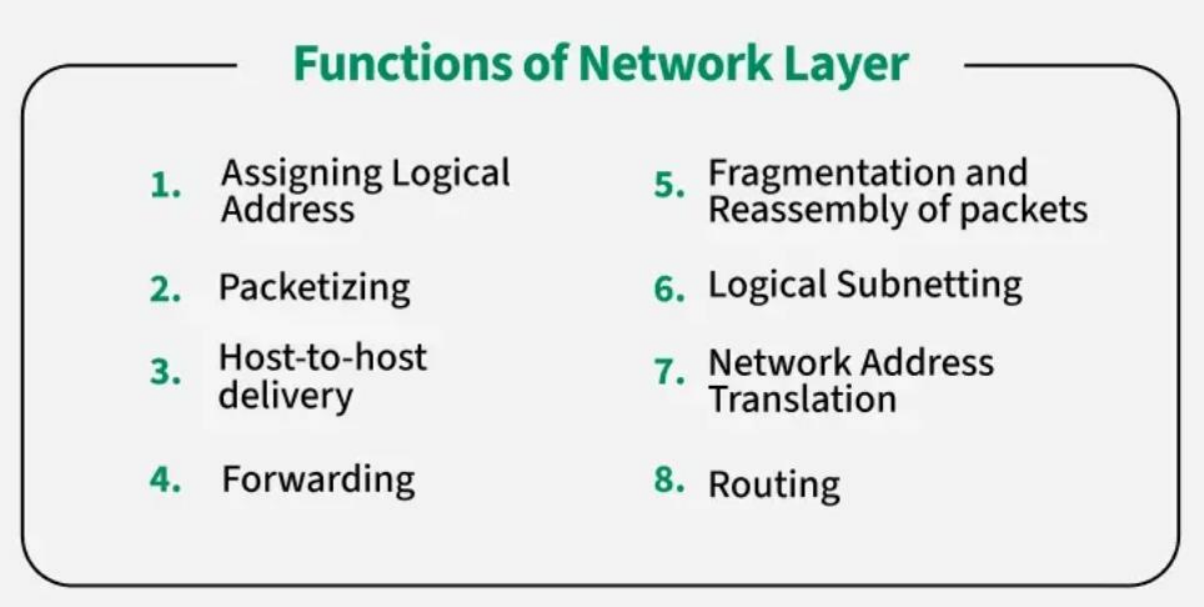
- **Services provided to Network Layer**

The network layer is a part of the communication process in computer networks. Its main job is to move data packets between different networks. It helps route these packets from the sender to the receiver across multiple paths and networks. Network-to-network connections enable the Internet to function. These connections happen at the network layer which sends data packets between different networks.

In the 7-layer OSI model, the network layer is layer 3.
The Internet Protocol (IP) is a key protocol used at this layer, along with other protocols for routing, testing, and encryption.

Services Offered by Network Layer

The **services** which are offered by the network layer are as follows:



1. Assigning Logical Address

Logical addressing is the process of assigning unique IP addresses (IPv4 or IPv6) to devices within a network. Unlike physical addresses (MAC addresses), logical addresses can change based on network configurations. These addresses are hierarchical and help identify both the network and the device within that network. Logical addressing is important for:

- Enabling communication between devices on different networks.
- Facilitating routing by providing location-based information.

2. Packetizing

The process of encapsulating the data received from the upper layers of the network (also called payload) in a network layer packet at the source and decapsulating the payload from the network layer packet at the destination is known as packetizing.

The source host adds a header that contains the source and destination address and some other relevant information required by the network layer protocol to the payload received from the upper layer protocol and delivers the packet to the data link layer.

The destination host receives the network layer packet from its data link layer, decapsulates the packet, and delivers the payload to the corresponding upper layer protocol. The routers in the path are not allowed to change either the source or the destination address. The routers in the path are not allowed to decapsulate the packets they receive unless they need to be fragmented.

3. Host-to-Host Delivery

The network layer ensures data is transferred from the source device (host) to the destination device (host) across one or multiple networks. This involves:

- Determining the destination address.
- Ensuring that data is transmitted without duplication or corruption.

Host-to-host delivery is a foundational aspect of communication in large-scale, interconnected systems like the internet.

4. Forwarding

Forwarding is the process of transferring packets between network devices such as routers, which are responsible for directing the packets toward their destination. When a router receives a packet from one of its attached networks, it needs to forward the packet to another attached network (unicast routing) or to some attached networks (in the case of multicast routing).The router uses:

- **Routing tables**: These tables store information about possible paths to different networks.

- **Forwarding decisions**: Based on the destination IP address in the packet header. Forwarding ensures that packets move closer to their destination efficiently.

5. Fragmentation and Reassembly of Packets

Some networks have a **maximum transmission unit (MTU)** that defines the largest packet size they can handle. If a packet exceeds the MTU, the network layer:

- **Fragments** the packet into smaller pieces.
- Adds headers to each fragment for identification and sequencing. At the destination, the fragments are **reassembled** into the original packet. This ensures compatibility with networks of varying capabilities without data loss.

6. Logical Subnetting

Logical subnetting involves dividing a large IP network into smaller, more manageable sub-networks (subnets). Subnetting helps:

- Improve network performance by reducing congestion.
- Enhance security by isolating parts of a network.
- Simplify network management and troubleshooting. Subnetting uses **subnet masks** to define the range of IP addresses within each subnet, enabling efficient address allocation and routing.

7. Network Address Translation (NAT)

NAT allows multiple devices in a private network to share a single public IP address for internet access. This is achieved by:

- Translating private IP addresses to a public IP address for outbound traffic.
- Reversing the process for inbound traffic. Benefits of NAT include:
- Conserving IPv4 addresses by reducing the need for unique public IPs for each device.
- Enhancing security by masking internal IP addresses from external networks.

8. Routing

Routing is the process of moving data from one device to another device. These are two other services offered by the network layer. In a network, there are a number of routes available from the source to the destination. The network layer specifies some strategies which find out the best possible route. This process is referred to as routing. There are a number of routing protocols that are used in this process and they should be run to help the routers coordinate with each other and help in establishing communication throughout the network.

Advantages

- Packetization service in the network layer provides ease of transportation of the data packets.
- Packetization also eliminates single points of failure in data communication systems.
- Routers present in the network layer reduce network traffic by creating collision and broadcast domains.
- With the help of Forwarding, data packets are transferred from one place to another in the network.

Disadvantages

- There is a lack of flow control in the design of the network layer.
- Congestion occurs sometimes due to the presence of too many datagrams in a network that is beyond the capacity of the network or the routers. Due to this, some routers may drop some of the datagrams, and some important pieces of information may be lost.
- Although indirect error control is present in the network layer, there is a lack of proper error control mechanisms as due to the presence of fragmented data packets, error control becomes difficult to implement.

➢ Elementary Data Link

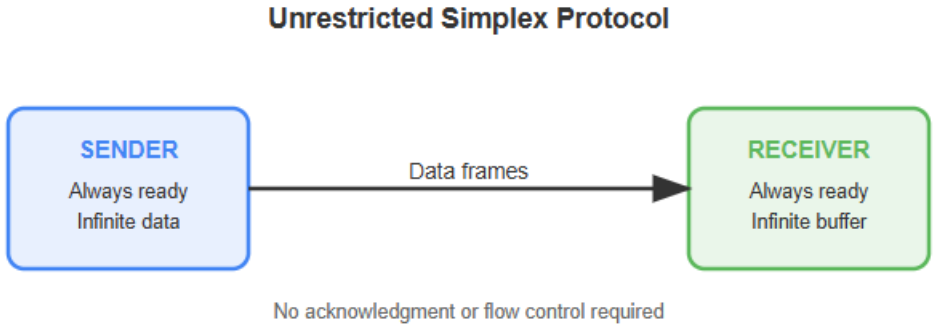
Elementary data link layer protocols are fundamental communication protocols that establish reliable data transmission between directly connected nodes. These protocols handle frame delivery, error detection, and flow control at the data link layer of the network stack.

Elementary data link protocols are classified into three categories based on their complexity and capabilities:

- **Protocol 1** − Unrestricted simplex protocol
- **Protocol 2** − Simplex stop and wait protocol
- **Protocol 3** − Simplex protocol for noisy channels

Unrestricted Simplex Protocol

The unrestricted simplex protocol represents the most basic form of data transmission. Data flows in one direction only, from sender to receiver. This protocol assumes ideal conditions where both transmission and receiving entities are always ready, processing time is negligible, infinite buffer space is available, and no errors occur.



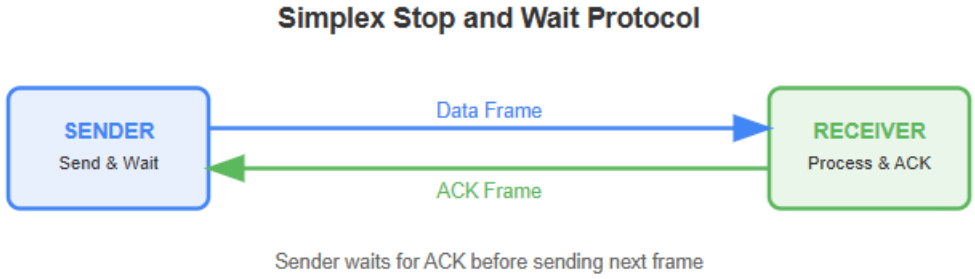
Simplex Stop and Wait Protocol

The simplex stop and wait protocol addresses the limitation of receiver processing speed. While data still flows in one direction, this protocol introduces flow control to prevent the sender from overwhelming the receiver.

The protocol operates through the following steps:

- **Step 1** − Sender transmits a data frame
- **Step 2** − Sender waits for acknowledgment (ACK) from receiver
- **Step 3** − Receiver processes the frame and sends ACK
- **Step 4** − Upon receiving ACK, sender transmits the next frame

This approach ensures that the sender cannot flood the receiver with frames faster than they can be processed.



Simplex Protocol for Noisy Channel

The **simplex protocol for noisy channels** is the most sophisticated of the three elementary protocols. It handles real-world conditions where data frames and acknowledgments can be corrupted or lost during transmission.

Key features include:

- **Sequence numbers** − Each frame carries a unique sequence number for identification
- **Timer mechanism** − Sender starts a timer after transmitting each frame
- **Retransmission** − If ACK is not received before timer expires, frame is retransmitted
- **Error detection** − Protocol can detect corrupted frames and acknowledgments

When acknowledgments are corrupted or data frames are damaged, the sender waits for a predetermined timeout period before retransmitting the frame.

Comparison of Elementary Protocols

Feature	Protocol 1	Protocol 2	Protocol 3
Flow Control	None	Stop and Wait	Stop and Wait
Error Handling	None	None	Detection & Retransmission
Acknowledgments	No	Yes	Yes
Sequence Numbers	No	No	Yes

- Layer protocols

Network Layer is responsible for the transmission of data or communication from one host to another host connected in a network. Rather than describing how data is transferred, it implements the technique for efficient transmission. In order to provide efficient communication protocols are used at the network layer. The data is being grouped into packets or in the case of extremely large data it is divided into smaller sub packets. Each protocol used has specific features and advantages. The below article covers in detail the protocols used at the network layer.

Functions of Network Layer

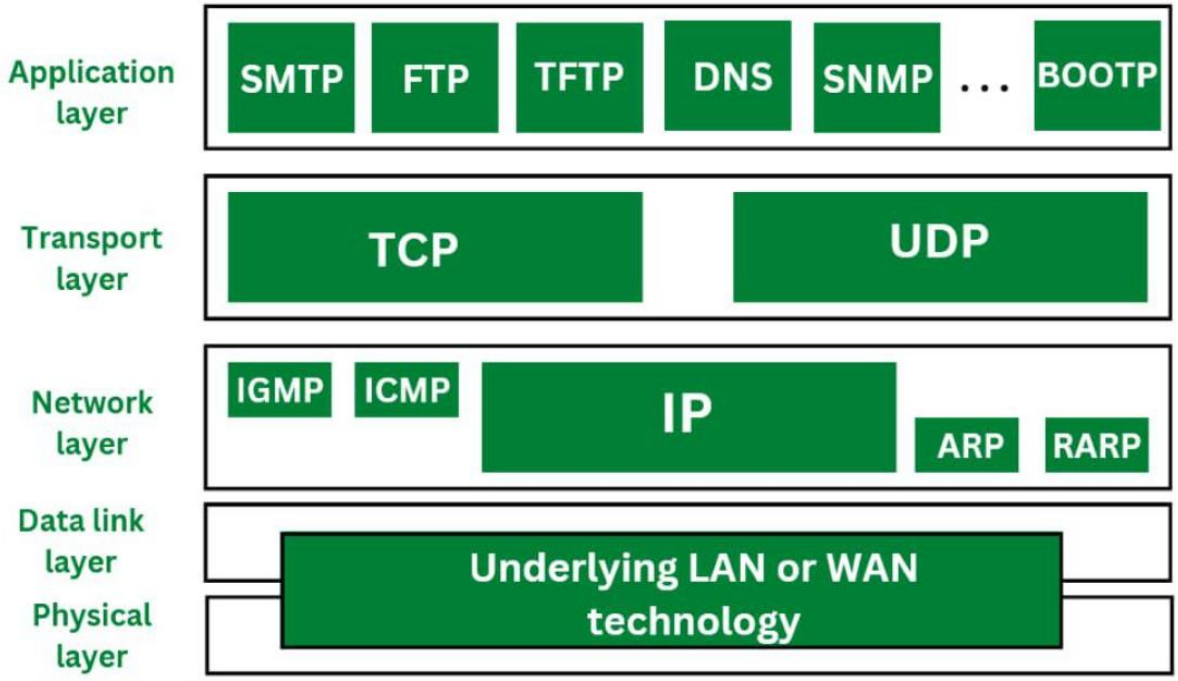
The network layer is responsible for providing the below-given tasks:

- **Logical Addressing:** Each device on the network needs to be identified uniquely. Therefore network layer provides an addressing scheme to identify the device. It places the IP address of every sender and the receiver in the header. This header consists of the network ID and host ID of the network.
- **Host-to-host Delivery of Data:** The network layer ensures that the packet is being delivered successfully from the sender to the receiver. This layer makes sure that the packet reaches the intended recipient only.
- **Fragmentation:** In order to transmit the larger data from sender to receiver, the network layer fragments it into smaller packets. Fragmentation is required because every node has its own fixed capacity for receiving data.
- **Congestion Control:** Congestion is defined as a situation where the router is not able to route the packets property which results in aggregation of packets in the network. Congestion occurs when a large amount of packets are flooded in the network. Therefore network layer controls the congestion of data packets in the network.
- **Routing and Forwarding:** Routing is the process that decides the route for transmission of packets from sender to receiver. It mostly chooses the shortest path between the sender and

the receiver. Routing protocols that are mostly used are path vector, distance vector routing, link state routing, etc.

Network Layer Protocols

There are various protocols used in the network layer. Each protocol is used for a different task. Below are the protocols used in the network layer:



1. IP (Internet Protocol)

IP stands for Internet Protocol. Internet Protocol helps to uniquely identify each device on the network. Internet protocol is responsible for transferring the data from one node to another node in the network. Internet protocol is a connectionless protocol therefore it does not guarantee the delivery of data. For the successful delivery higher level protocols such as TCP are used to guarantee the data transmission. The Internet Protocol is divided in two types. They are:

- **IPv4:** IPv4 provides with the 32 bit address scheme. IPv4 addressing has four numeric fields and are separated by dot. IPv4 can be configured either using DHCP or manually. IPv4 does not provide with more security features as it does not support authentication or encryption techniques. IPv4 is further divided into five classes as Class A, Class B, Class C, Class D and Class E.
- **IPv6:** IPv6 is the most recent version of IP. If provided with a 128 bit addressing scheme. IP address has eight fields that are separated by colon, and these fields are alphanumeric. The IPv6 address is represented in hexadecimal. IPv6 provides with more security features such as authentication and encryption. IPv6 supports end-to-end connection integrity. IPv6 provides with more range of IP address as compared to IPv4.

2. ARP (Address Resolution Protocol)

ARP stands for Address Resolution Protocol. ARP is used to convert the logical address ie. IP address into physical address ie. MAC address. While communicating with other nodes, it is necessary to know the MAC address or physical address of the destination node. If any of the node in a network wants to know the physical address of another node in the same network, the host then sends an ARP query packet. This ARP query packet consists of IP address and MAC address of source host and only the IP address of destination host. This ARP packet is then received to every node present in the network. The node with its own IP address recognises it and sends it MAC address to the requesting node. But sending and receiving such packets to know the MAC address of destination node it increases the traffic load. Therefore in order to reduce this traffic

and improve the performance, the systems that makes use of ARP maintain a cache of recently acquired IP into MAC address bindings.

3. RARP

RARP stands for Reverse Address Resolution Protocol. RARP works opposite of ARP. Reverse Address Resolution Protocol is used to convert MAC address ie. physical address into IP address ie. logical address. RARP provides with a feature for the systems and applications to get their own IP address from a DNS(Domain Name System) or router. This type of resolution is required for various tasks such as executing reverse DNS lookup. As Reverse Address Resolution Protocol works at low level it requires direct network addresses. The reply from the server mostly carries a small information but the 32 bit internet address is used and it does not exploit the full potential of a network such as ethernet.

4. ICMP

ICMP stands for Internet Control Message Protocol. ICMP is a part of IP protocol suite. ICMP is an error reporting and network diagnostic protocol. Feedback in the network is reported to the designated host. Meanwhile, if any kind of error occur it is then reported to ICMP. ICMP protocol consists of many error reporting and diagnostic messages. ICMP protocol handles various kinds of errors such as time exceeded, redirection, source quench, destination unreachable, parameter problems etc. The messages in ICMP are divided into two types. They are given below:

- Error Message: Error message states about the issues or problems that are faced by the host or routers during processing of IP packet.
- Query Message: Query messages are used by the host in order to get information from a router or another host.

5. IGMP

IGMP stands for Internet Group Message Protocol. IGMP is a multicasting communication protocol. It utilizes the resources efficiently while broadcasting the messages and data packets. IGMP is also a protocol used by TCP/IP. Other hosts connected in the network and routers makes use of IGMP for multicasting communication that have IP networks. In many networks multicast routers are used in order to transmit the messages to all the nodes. Multicast routers therefore receives large number of packets that needs to be sent. But to broadcast this packets is difficult as it would increase the overall network load. Therefore IGMP helps the multicast routers by addressing them while broadcasting. As multicast communication consists of more than one senders and receivers the Internet Group Message Protocol is majorly used in various applications such as streaming media, web conference tools, games, etc.

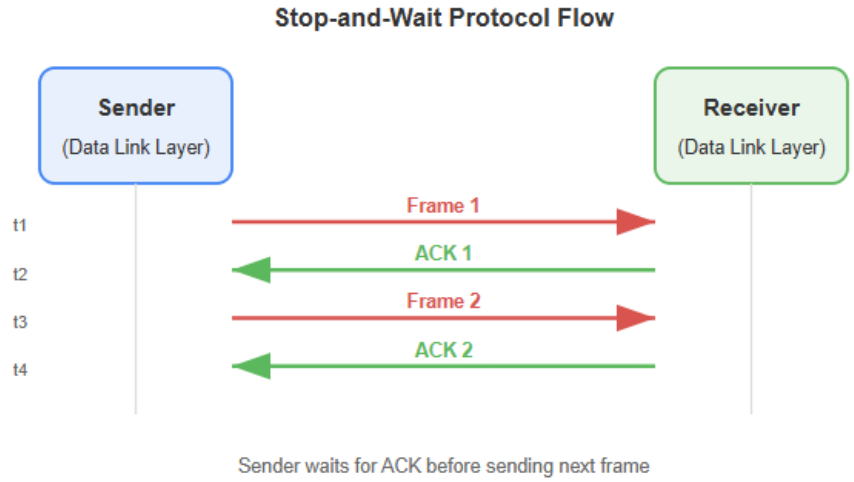
- Simplex protocol, Simplex stop and wait, Simplex protocol for a noisy channel.

Stop-and-Wait protocol is a **data link layer protocol** for transmission of frames over noiseless channels. It provides unidirectional data transmission with flow control facilities but without error control facilities.

This protocol takes into account the fact that the receiver has a finite processing speed. If data frames arrive at the receiver's end at a rate greater than its processing rate, frames will be dropped. To avoid this, the receiver sends an **acknowledgement** for each frame upon arrival. The sender transmits the next frame only after receiving a positive acknowledgement confirming the receiver is ready for further data processing.

How It Works

- **Sender Site** − The data link layer waits for the network layer to provide a data packet. It checks whether it can send the frame, and if the physical layer gives positive notification, it creates frames from the data and sends them. It then waits for acknowledgement before sending the next frame.
- **Receiver Site** − The data link layer waits for a frame to arrive. When it arrives, the receiver processes it, delivers it to the network layer, and sends an acknowledgement back to the sender.



Advantages and Disadvantages

Advantages	Disadvantages
Simple implementation and flow control	Low bandwidth utilization due to waiting
Prevents receiver buffer overflow	No error detection or correction
Reliable for error-free channels	Inefficient for high-speed networks

➢ Simplex protocol for a noisy channel.

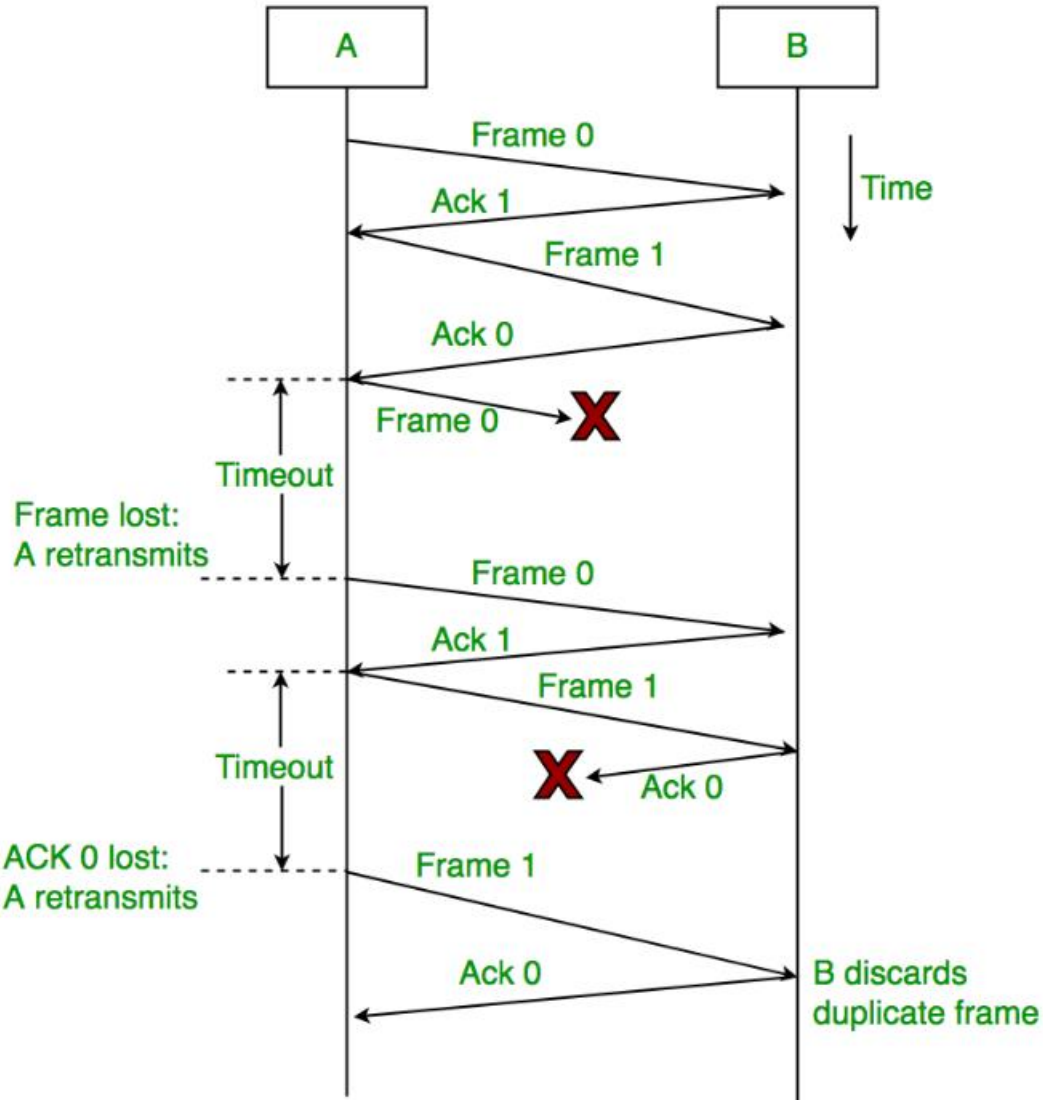
Noisy channel protocols are commonly known as communication protocols. These protocols are designed to ensure a reliable data transmission over a channel in which there is a high probability of errors or data loss. In computer networks, these protocols are typically used to overcome the effects of channel noise during data transmission which can result from electromagnetic interference, attenuation etc. that cause signal degradation. So, noisy channel protocols perform an important role to ensure reliable communication in computer networks, especially in environments where channel noise is a significant concern.

Example 1

A very known example of a noisy channel protocol is the Automatic Repeat Request (ARQ) protocol. ARQ protocol is basically used in many network protocols, such as TCP. It ensures reliable data transmission. The workflow ARQ is very simple as it sends data packets and requests acknowledgement of receipt from the receiver end. If the sender does not receive an acknowledgement from the receiver within a specified time period, it assumes that the packet was lost or corrupted and resends it.

Stop-and-Wait Automatic Repeat Request

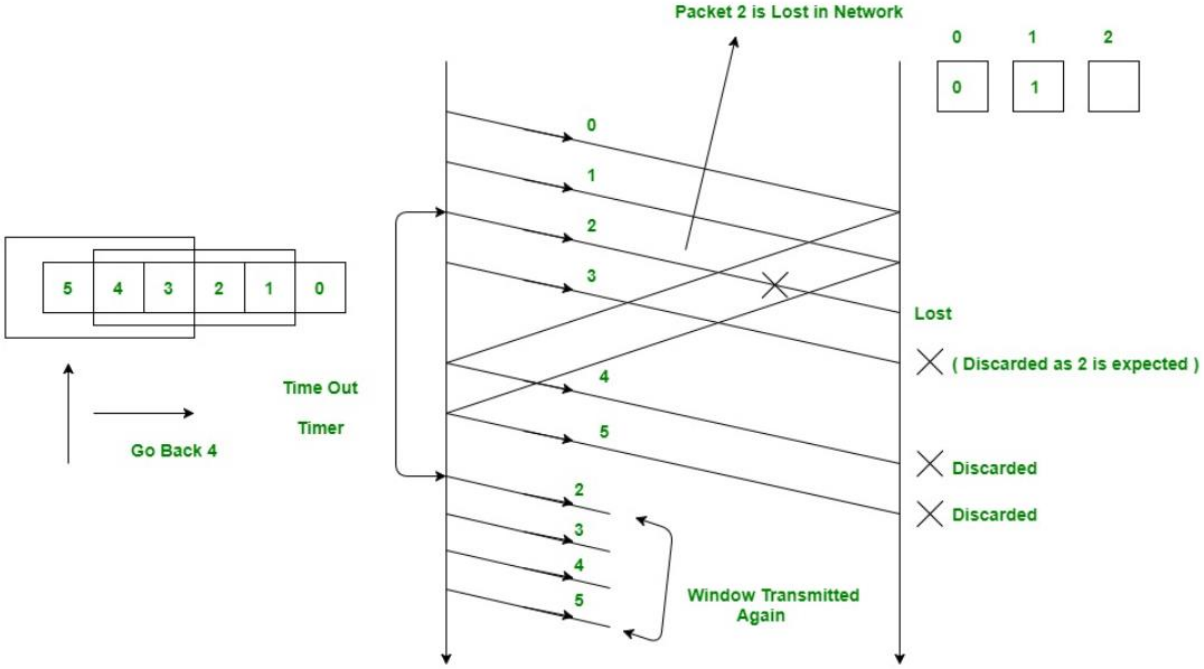
Stop-and-Wait Automatic Repeat Request(ARQ) is a common type of error control protocol widely used in computer networks to ensure reliable data transmission. This protocol works by transmitting a single data packet and waiting for an acknowledgement before sending the next packet. In this ARQ protocol, the sender sends a data packet to the receiver and waits for an acknowledgement from the receiver side. The receiver after receiving the packet, sends an acknowledgement back to the sender indicating that the packet has been successfully received. If the sender does not receive an acknowledgement from the receiver side within a specified time frame then the sender retransmits the packet. So, this protocol is called "stop-and-wait" because the sender must wait for the acknowledgement before sending the next packet. Roe that reason the protocol became relatively slow, as it does not allow for pipelining or sending multiple packets at once. This protocol is commonly used in such situations where data loss is not frequent or where the cost of retransmitting a lost packet is low.



Go-Back-N Automatic Repeat Request

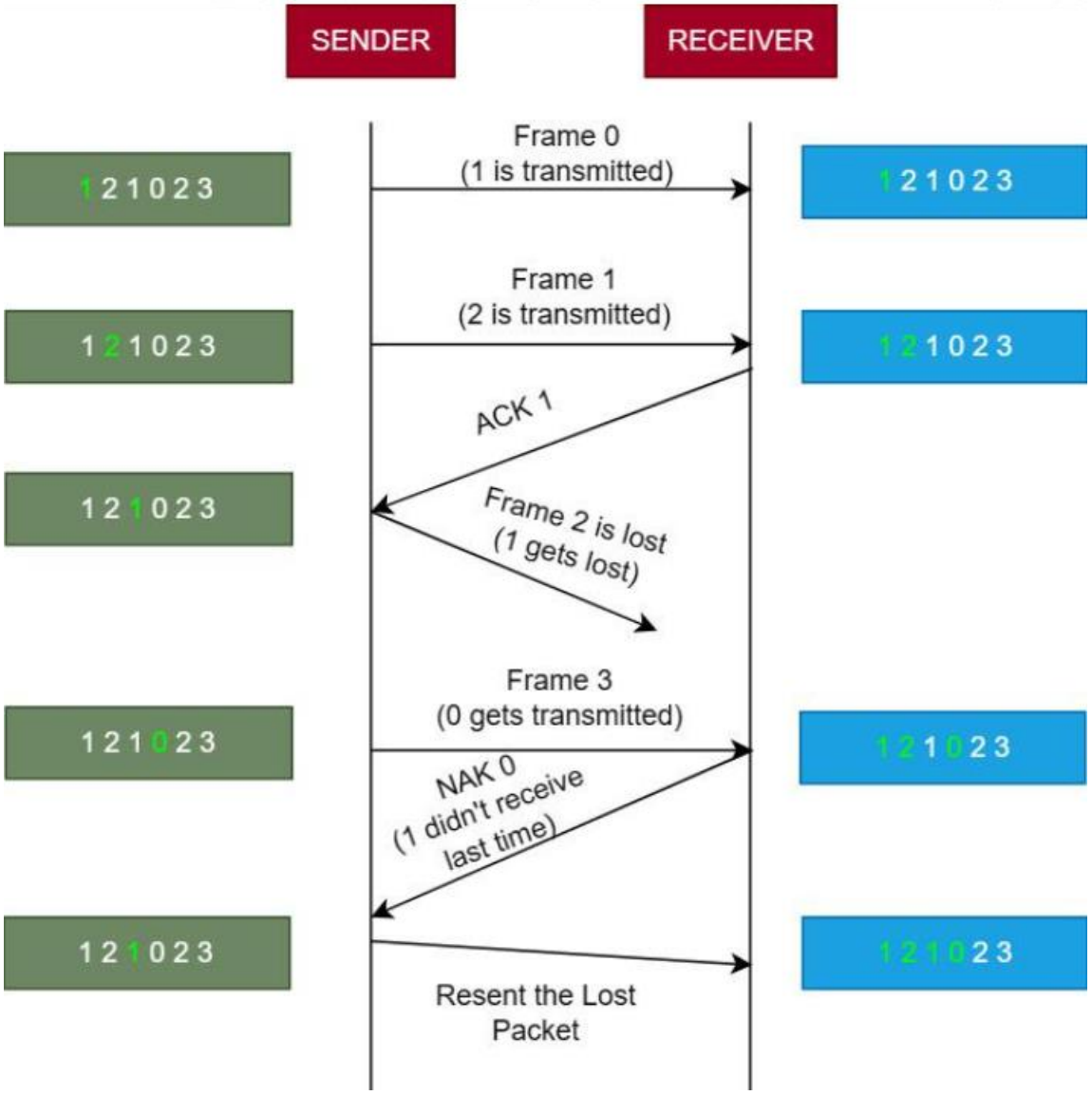
Go-Back-N Automatic Repeat Request(ARQ) is also an error control protocol used in computer networks to ensure reliable data transmission. It this protocol the sender is allowed to transmit multiple packets before waiting for an acknowledgement which improves the efficiency of data transmission. In this protocol the sender sends multiple numbers of packets to the receiver without waiting for an acknowledgement for each packet. The receiver receives the data packets and sends an acknowledgement for the last correctly received packet only. If the sender does not

receive an acknowledgement from the receiver end within a specified time frame then sender retransmits all packets starting from the last unacknowledged packet. So, this protocol is called "go-back-n" because the sender must go back and retransmit all the packets that were not acknowledged. This means that if a single packet is lost then all the subsequent packets must be retransmitted and this can lead to inefficient use of network resources. This protocol is commonly used in situations where data loss is moderate and network resources are not limited.



Selective Repeat Automatic Repair Request -

The selective repeat ARQ is like the Go-Back-N ARQ, which lets the user to transmit multiple data packets before they receive any acknowledgment from the server side. But, there is a small difference in this mechanism, unlike the Go-Back-N ARQ, which informs the user to retransmit the entire window of packets sent to it, if any one packet is found erroneous. The Selective Repeat ARQ doesn't inform the user to retransmit the entire window, as it checks each of the packets one by one and if found any error, then it only retransmits only those faulty packages, not the entire window. In this protocol, the client (packet sender) maintains a timer for each window of packets, now after the successful transmission of packets, if the sender doesn't receive the ACK package in between a certain time, then the sender retransmits those packets only. Now on the receiver side, if the receiver receives the expected packet properly then it sends back an ACK packet to the sender for the next packet. But if there is any error, then the receiver sends a NAK (Negative Acknowledgement) packet to the sender and asks to retransmit that packet only.



- **Sliding window protocol**
  - ✓ **One-Bit Sliding Window Protocol (Stop-and-Wait Protocol)**

The One-Bit Sliding Window Protocol is the simplest sliding window protocol used for reliable data transmission. It is also known as the Stop-and-Wait ARQ (Automatic Repeat reQuest) protocol.

Key Features

- Sender window size = 1
- Receiver window size = 1
- Uses 1-bit sequence numbers (0 and 1).
- The sender sends one frame at a time and waits for its acknowledgment (ACK) before sending the next frame.

Working

1. The sender sends a frame with sequence number 0.
2. The receiver receives the frame, checks for errors, and sends ACK 1 (indicating it expects the next frame with sequence number 1).
3. The sender receives ACK 1 and sends the next frame with sequence number 1.
4. The receiver sends ACK 0 after receiving frame 1.
5. The process repeats with sequence numbers alternating between 0 and 1.

Example

Step	Sender	Receiver	ACK Sent
1	Frame 0	Receives Frame 0	ACK 1
2	Frame 1	Receives Frame 1	ACK 0
3	Frame 0	Receives Frame 0	ACK 1

Handling Lost Frames

- If a frame is lost, the sender does not receive an ACK before the timeout expires.
- The sender retransmits the same frame.
- If an ACK is lost, the sender retransmits the previous frame after timeout.
- The receiver identifies duplicate frames using the 1-bit sequence number and discards duplicates while resending the appropriate ACK.

Advantages

- Simple and easy to implement.
- Ensures reliable communication.
- Detects duplicate frames.

Disadvantages

- Low channel utilization because the sender waits after every frame.
- Not suitable for high-speed or long-delay networks.

1. Stop-and-Wait Protocol (One-Bit Sliding Window Protocol)

Definition

The Stop-and-Wait Protocol is the simplest flow control protocol in which the sender transmits one frame at a time and waits for an acknowledgment (ACK) before sending the next frame.

Working

1. Sender sends one frame.
2. Receiver checks for errors.
3. If the frame is correct, receiver sends an ACK.
4. Sender sends the next frame only after receiving the ACK.
5. If the ACK is not received before timeout, the sender retransmits the frame.

Characteristics

- Sender window size = 1
- Receiver window size = 1
- Uses 1-bit sequence numbers (0 and 1)

Advantages

- Simple to implement.
- Reliable transmission.
- Error detection and retransmission.

Disadvantages

- Low channel utilization.
- Poor performance over long-distance networks.

2. Go-Back-N ARQ (Automatic Repeat Request)

Definition

Go-Back-N ARQ is a sliding window protocol where the sender can send multiple frames without waiting for acknowledgments, but if an error occurs, it retransmits the erroneous frame and all subsequent frames.

Window Size

- Sender window size = N
- Receiver window size = 1

Working

Suppose the sender sends Frames:

0 1 2 3 4

If Frame 2 is lost:

- Receiver receives Frame 0 ✓
- Receiver receives Frame 1 ✓
- Receiver expects Frame 2
- Receiver discards Frames 3 and 4
- Sender retransmits Frames 2, 3, and 4

Diagram

Sender Receiver

Frame0 ---------->
<----------ACK1----------

Frame1 ---------->
<----------ACK2----------

Frame2 ----X (Lost)

Frame3 ----------> Discard
Frame4 ----------> Discard

Timeout

Frame2 ---------->
Frame3 ---------->
Frame4 ---------->

<----------ACK5----------

Advantages

- Better throughput than Stop-and-Wait.
- Efficient when transmission errors are rare.

Disadvantages

- Retransmits many correctly received frames.
- Bandwidth is wasted during retransmission.

3. Selective Repeat ARQ

Definition

Selective Repeat ARQ is an advanced sliding window protocol in which only the erroneous or lost frames are retransmitted.

Window Size

- Sender window size = N
- Receiver window size = N

Working
Sender transmits
0 1 2 3 4
If Frame 2 is lost
Receiver receives
0 ✓
1 ✓
2 Lost
3 Stored
4 Stored
Receiver requests only Frame 2.
Sender retransmits only Frame 2.
Receiver then delivers frames in order.
Diagram
Sender Receiver

Frame0 ---------->
<----------ACK0----------

Frame1 ---------->
<----------ACK1----------

Frame2 ----X Lost

Frame3 ----------> Buffer
<----------ACK3----------

Frame4 ----------> Buffer
<----------ACK4----------

Timeout

Frame2 ---------->

Receiver now delivers
2 3 4
Advantages

- Efficient bandwidth utilization.
- Retransmits only lost frames.
- Higher throughput.

Disadvantages

- More complex implementation.
- Requires receiver buffering.
- More memory is needed.

Comparison of ARQ Protocols

Feature	Stop-and-	Go-Back-N	Selective Re
Sender Window	1	N	N
Receiver Windo	1	1	N
Retransmission	One fram	Lost frame and following frames	Only lost fra
Complexity	Low	Medium	High
Throughput	Low	Medium	High
Receiver Buffer	No	No	Yes

4. HDLC (High-Level Data Link Control)

Definition

HDLC is a bit-oriented Data Link Layer protocol used for reliable communication over point-to-point and multipoint links.

Features

- Bit-oriented protocol
- Full-duplex communication
- Error detection using CRC
- Flow control
- Sliding window mechanism
- Supports reliable transmission

HDLC Frame Format

Flag | Address | Control | Information | FCS | Flag

8-bit 8-bit 8-bit Variable 16/32 8-bit

Fields

1. Flag

- Value = 01111110
- Indicates beginning and end of frame.

2. Address

Contains receiver address.

3. Control

Contains

- Sequence numbers
- ACK information
- Frame type

4. Information

Actual user data.

5. FCS (Frame Check Sequence)

CRC used for error detection.

HDLC Frame Types

1. Information (I-frame)

2. Supervisory (S-frame)
3. Unnumbered (U-frame)

Advantages

- Reliable communication.
- Supports error and flow control.
- High efficiency.

Disadvantages

- More complex than PPP.
- Higher processing overhead.

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5. Point-to-Point Protocol (PPP)

Definition

PPP is a byte-oriented Data Link Layer protocol used to establish direct communication between two network devices over serial links.

Applications

- Dial-up Internet
- DSL
- Broadband connections
- Router-to-router communication

PPP Frame Format

Flag | Address | Control | Protocol | Data | FCS | Flag

Fields

Flag

Start and end delimiter.

Address

Usually 11111111.

Control

Usually 00000011.

Protocol

Identifies the network layer protocol (IPv4, IPv6, etc.).

Data

Actual user data.

FCS

Error detection using CRC.

PPP Components

1. LCP (Link Control Protocol)

- Establishes the link.
- Configures parameters.
- Tests the link.

2. NCP (Network Control Protocol)

- Configures network-layer protocols.
- Supports IPv4, IPv6, etc.

Authentication Methods

- PAP (Password Authentication Protocol)

- CHAP (Challenge Handshake Authentication Protocol)

Advantages of PPP

- Supports authentication.
- Error detection using CRC.
- Easy to configure.
- Supports multiple network-layer protocols.

Disadvantages of PPP

- Supports only point-to-point communication.
- Does not provide error correction (only error detection).
- Less suitable for modern high-speed LANs.

HDLC vs PPP

Feature	HDLC	PPP
Type	Bit-oriented	Byte-oriented
Communication	Point-to-point & multipoin	Point-to-point only
Authentication	Not built-in	PAP and CHAP supported
Error Detection	CRC	CRC
Error Recovery	Yes	No (only detection)
Configuration	More complex	Simpler
Common Use	WAN links	Internet and serial links